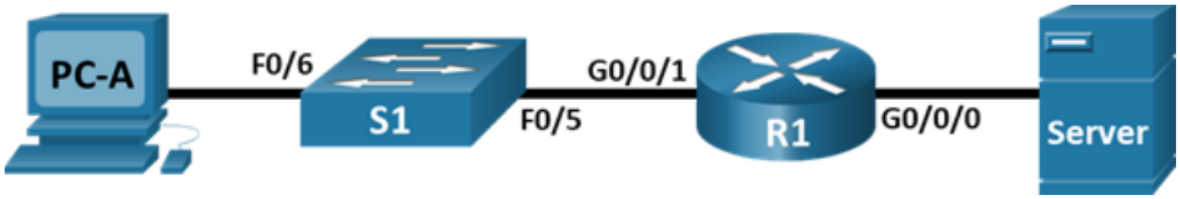
1.6.2 Packet Tracer - Configure Basic Router Settings

Physical Mode Lab Report

Topology



Network topology: PC-A -> S1 -> R1 -> Server

Addressing Table

Device	Interface	IP Address / Prefix	Default Gateway
R1	G0/0/0	192.168.0.1 /24 2001:db8:acad::1 /64 fe80::1	N/A
R1	G0/0/1	192.168.1.1 /24 2001:db8:acad:1::1 /64 fe80::1	N/A
R1	Loopback0	10.0.0.1 /24 2001:db8:acad:2::1 /64 fe80::1	N/A
PC-A	NIC	192.168.1.10 /24 2001:db8:acad:1::10 /64	192.168.1.1 fe80::1
Server	NIC	192.168.0.10 /24 2001:db8:acad::10 /64	192.168.0.1 fe80::1

Original addressing table image:

Device	Interface	IP Address / Prefix	Default Gateway
R1	G0/0/0	192.168.0.1 /24	N/A
		2001:db8:acad::1 /64	
		fe80::1	
	G0/0/1	192.168.1.1 /24	
		2001:db8:acad:1::1 /64	
		fe80::1	
	Loopback0	10.0.0.1 /24	
		2001:db8:acad:2::1 /64	
		fe80::1	
PC-A	NIC	192.168.1.10 /24	192.168.1.1
		2001:db8:acad:1::10 /64	fe80::1
Server	NIC	192.168.0.10 /24	192.168.0.1
		2001:db8:acad::10 /64	fe80::1

R1 Configuration Commands

Use these commands on R1 after cabling PC-A RS232 to the router Console port.

```
enable
configure terminal
hostname R1
ip domain-name ccna-lab.com
service password-encryption
security passwords min-length 12
username SSHadmin secret 55Hadm!n2020
crypto key generate rsa general-keys modulus 1024
enable secret $cisco!PRIV*

line console 0
password $cisco!!CON*
exec-timeout 4 0
login
exit

line vty 0 4
password $cisco!!VTY*
transport input ssh
exec-timeout 4 0
login local
exit

banner motd #Unauthorized access is prohibited.#
login block-for 120 attempts 3 within 60
ipv6 unicast-routing

interface gigabitEthernet0/0/0
description Link to Server
```

```
ip address 192.168.0.1 255.255.255.0
ipv6 address 2001:db8:acad::1/64
ipv6 address fe80::1 link-local
no shutdown
exit

interface gigabitEthernet0/0/1
description Link to S1 / PC-A LAN
ip address 192.168.1.1 255.255.255.0
ipv6 address 2001:db8:acad:1::1/64
ipv6 address fe80::1 link-local
no shutdown
exit

interface loopback0
description Loopback interface
ip address 10.0.0.1 255.255.255.0
ipv6 address 2001:db8:acad:2::1/64
ipv6 address fe80::1 link-local
exit

end
clock set 18:00:00 25 Jun 2026
copy running-config startup-config
```

Verification Commands

```
show ip interface brief
show ipv6 interface brief
show ip route
show version
show startup-config
show running-config | section vty
show version | include register
```

Task Questions - Answers

1. **What would be the result of reloading the router prior to completing the copy running-config startup-config command?**

   Any changes made in the running configuration would be lost. The router would reload using the last saved startup configuration.

2. **Were the pings from PC-A to the Server successful?**

   Yes. After PC-A, Server, and R1 are configured correctly, both IPv4 and IPv6 pings from PC-A to Server should be successful.

3. **Was remote access to R1 using the Loopback IPv4 address successful?**

   Yes. SSH access to the R1 Loopback0 IPv4 address should be successful when using username SSHadmin and password 55Hadm!n2020.

4. **Was remote access to R1 using the Loopback IPv6 address successful?**

   Yes. SSH access to the R1 Loopback0 IPv6 address should be successful when using username SSHadmin and password 55Hadm!n2020.

5. **Why is the Telnet protocol considered to be a security risk?**

   Telnet sends data, including usernames and passwords, in clear text. SSH is safer because it encrypts the remote session.

6. **What is the name of the IOS image that the router is running?**

   Use show version to verify the exact image. A common Packet Tracer Cisco 4321 ISR image is isr4300-universalk9.16.06.04.SPA.bin.

7. **How much NVRAM does the router have?**

   Use show version to verify the exact value. A common Packet Tracer Cisco 4321 ISR value is 32768K bytes of NVRAM.

8. **How much Flash memory does the router have?**

   Use show version to verify the exact value. A common Packet Tracer Cisco 4321 ISR value is about 3223551K bytes of flash memory.

9. **What would be the boot process if the configuration register was 0x2142?**

   The router would ignore the startup configuration in NVRAM during the next reload. This setting is commonly used for password recovery.

10. **How are passwords presented in the startup configuration output?**

    Passwords are shown in encrypted form. Plaintext line passwords are encrypted by service password-encryption, and secret passwords appear as hashed secret values.

11. **What is the result of using show running-config | section vty?**

    It filters the running configuration and displays only the VTY line configuration section.

12. **What code is used in the routing table to indicate a directly connected network?**

    The code C is used for directly connected networks.

13. **How many route entries are coded with a C code in the routing table?**

    There should be three C routes: 192.168.0.0/24, 192.168.1.0/24, and 10.0.0.0/24.

14. **What command changed the status of the Gigabit Ethernet ports from administratively down to up?**

    The no shutdown command changed the interfaces from administratively down to up.

15. **What is the meaning of the [up/up] part of the show ipv6 int brief output?**

    The first up means the physical interface is active. The second up means the line protocol is active.

16. **What is the IPv6 address assigned to Server after changing it from static IPv6?**

The server receives a SLAAC IPv6 address in the 2001:db8:acad::/64 network. In this activity, it is commonly 2001:db8:acad:0:207:ecff:fea5:6286, based on the Server MAC address. Verify with ipconfig on Server.

17. **What is the default gateway assigned to Server after changing it from static IPv6?**

    The IPv6 default gateway is fe80::1.

18. **From PC-B/PC-A, was the ping to the R1 default gateway link-local address successful?**

    Yes. The ping to the R1 link-local gateway address fe80::1 should be successful from a host on the same local link.

19. **From Server, was the ping to the R1 IPv6 unicast address 2001:db8:acad::1 successful?**

    Yes. The ping from Server to 2001:db8:acad::1 should be successful.

Reflection Questions - Answers

20. **If a technician suspects that an interface was not enabled, what show command could be used?**

    Use show ip interface brief. This command shows whether interfaces are up/up, down/down, or administratively down.

21. **If a technician suspects that an interface was assigned an incorrect subnet mask, what show command could be used?**

    Use show running-config. This command displays the IP address and subnet mask configured on each interface.

Original Task Pages

Tasks Page 1

Objectives

Part 1: Set Up the Topology and Initialize Devices

Part 2: Configure Devices and Verify Connectivity

Part 3: Display Router Information

Background / Scenario

This is a comprehensive Packet Tracer Physical Mode (PTPM) activity to review previously covered IOS router commands. In Parts 1 and 2, you will cable the equipment and complete basic configurations and interface settings on the router.

In Part 3, you will use SSH to connect to the router remotely and use the IOS commands to retrieve information from the device to answer questions about the router.

For review purposes, this activity provides the commands necessary for specific router configurations.

Instructions

Part 1: Set Up the Topology and Initialize Devices

Step 1: Cable the network as shown in the topology.

a. Click and drag the **Cisco 4321 ISR**, the **Cisco 2960 Switch**, and the **Server** from the **Shelf** to the **Rack**.

b. Click and drag the **PC** from the **Shelf** to the **Table**.

c. Cable the devices as specified in the topology diagram. Use **Copper Straight-through** cables for network connections.

d. From the **PC**, connect a **Console Cable** to the **Cisco 4321 ISR**.

e. Power on the **Cisco 4321 ISR**, **PC-A**, and **Server**. The power button for **Server** is on the bottom right. The 2960 switch should power on automatically.

Part 2: Configure Devices and Verify Connectivity

Step 1: Configure the PC interfaces.

a. Configure the IP address, subnet mask, and default gateway settings on **PC-A**.

b. Configure the IP address, subnet mask, and default gateway settings on **Server**.

Step 2: Configure the router.

a. Console into the router and enable privileged EXEC mode.

b. Enter configuration mode.

c. Assign a device name to the router.

d. Set the router's domain name as ccna-lab.com.

e. Encrypt the plaintext passwords.

f. Configure the system to require a minimum 12-character password.

g. Configure the username **SSHadmin** with an encrypted password of **55Hadm!n2020**.

h. Generate a set of crypto keys with a 1024 bit modulus.

i. Assign **$cisco!PRIV*** as the privileged EXEC password.

j. Assign **$cisco!!CON*** as the console password. Configure sessions to disconnect after four minutes of inactivity, and enable login.

k. Assign **$cisco!!VTY*** as the vty password. Configure the vty lines to accept SSH connections only. Configure sessions to disconnect after four minutes of inactivity, and enable login using the local database.

l. Create a banner that warns anyone accessing the device that unauthorized access is prohibited.

m. Enable IPv6 routing.

n. Configure all three interfaces on the router with the IPv4 and IPv6 addressing information from the addressing table above. Configure all three interfaces with descriptions. Activate all three interfaces.

The router should not allow vty logins for two minutes if three failed login attempts occur within 60 seconds.

o. Set the clock on the router.

p. Save the running configuration to the startup configuration file.

Tasks Page 2

What would be the result of reloading the router prior to completing the **copy running-config startup-config** command?

Step 3: Verify network connectivity.

a. Using the command line at **PC-A**, ping the IPv4 and IPv6 addresses for **Server**.

Were the pings successful?

b. From **PC-A**, remotely access **R1** using the Telnet / SSH client.

Using the Telnet / SSH client on PC-A, open an SSH session to the R1 Loopback interface IPv4 address. Ensure that the Connection Type is set to **SSH** and use **SSHadmin** as the username. When prompted, enter the password **55Hadm!n2020**.

Was remote access successful?

c. Using the Telnet / SSH client on **PC-A**, open an SSH session to the R1 Loopback interface IPv6 address. Ensure that the Connection Type is set to **SSH** and use **SSHadmin** as the username. When prompted, enter the password **55Hadm!n2020**.

Was remote access successful?

Why is the Telnet protocol considered to be a security risk?

Part 3: Display Router Information

In Part 3, you will use **show** commands from an SSH session to retrieve information from the router.

Step 1: Establish an SSH session to R1.

Using Telnet / SSH client on PC-A, open an SSH session to the R1 Loopback interface IPv6 address and log in as **SSHadmin** with the password **55Hadm!n2020**.

Step 2: Retrieve important hardware and software information.

a. Use the **show version** command to answer questions about the router.

What is the name of the IOS image that the router is running?

How much non-volatile random-access memory (NVRAM) does the router have?

How much Flash memory does the router have?

b. The **show** commands often provide multiple screens of outputs. Filtering the output lets you display certain sections of the output. To enable the filtering command, enter a pipe (**|**) character after a **show** command, followed by a filtering parameter and a filtering expression. You can match the output to the filtering statement by using the **include** keyword to display all lines from the output that contain the filtering expression. Filter the **show version** command, using **show version | include register** to answer the following question.

What would be the boot process for the router on the next reload if the configuration register was 0x2142?

Step 3: Display the startup configuration.

a. Use the **show startup-config** command on the router to answer the following question.

How are passwords presented in the output?

b. Use the **show running-config | section vty** command.

What is the result of using this command?

Step 4: Display the routing table on the router.

Use the **show ip route** command on the router to answer the following questions.

What code is used in the routing table to indicate a directly connected network?

How many route entries are coded with a C code in the routing table?

Step 5: Display a summary list of the interfaces on the router.

a. Use the **show ip interface brief** command on the router to answer the following question.

What command changed the status of the Gigabit Ethernet ports from administratively down to up?

b. Use the **show ipv6 int brief** command to verify IPv6 settings on R1.

What is the meaning of the [up/up] part of the output?

Tasks Page 3

c. On **Server**, change its configuration so that it no longer has a static IPv6 address. Then, issue the **ipconfig** command on **Server** to examine the IPv6 configuration.

What is the IPv6 address assigned to Server?

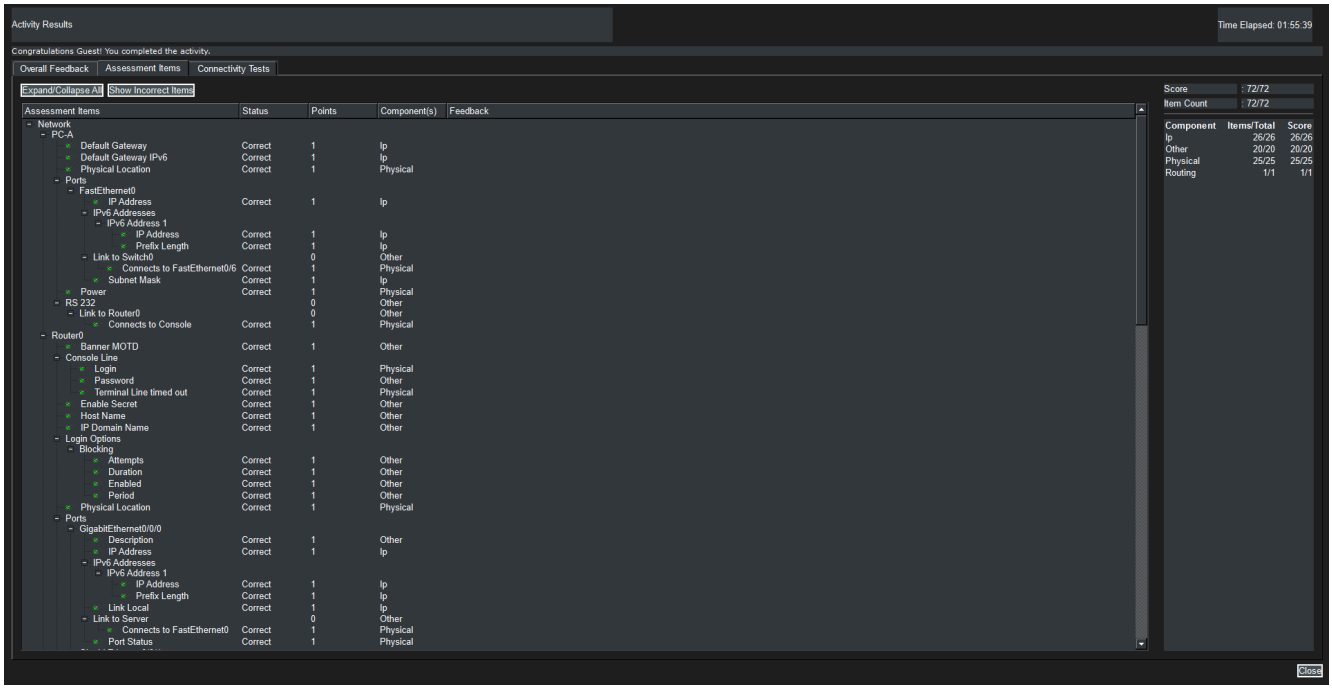
What is the default gateway assigned to **Server**?

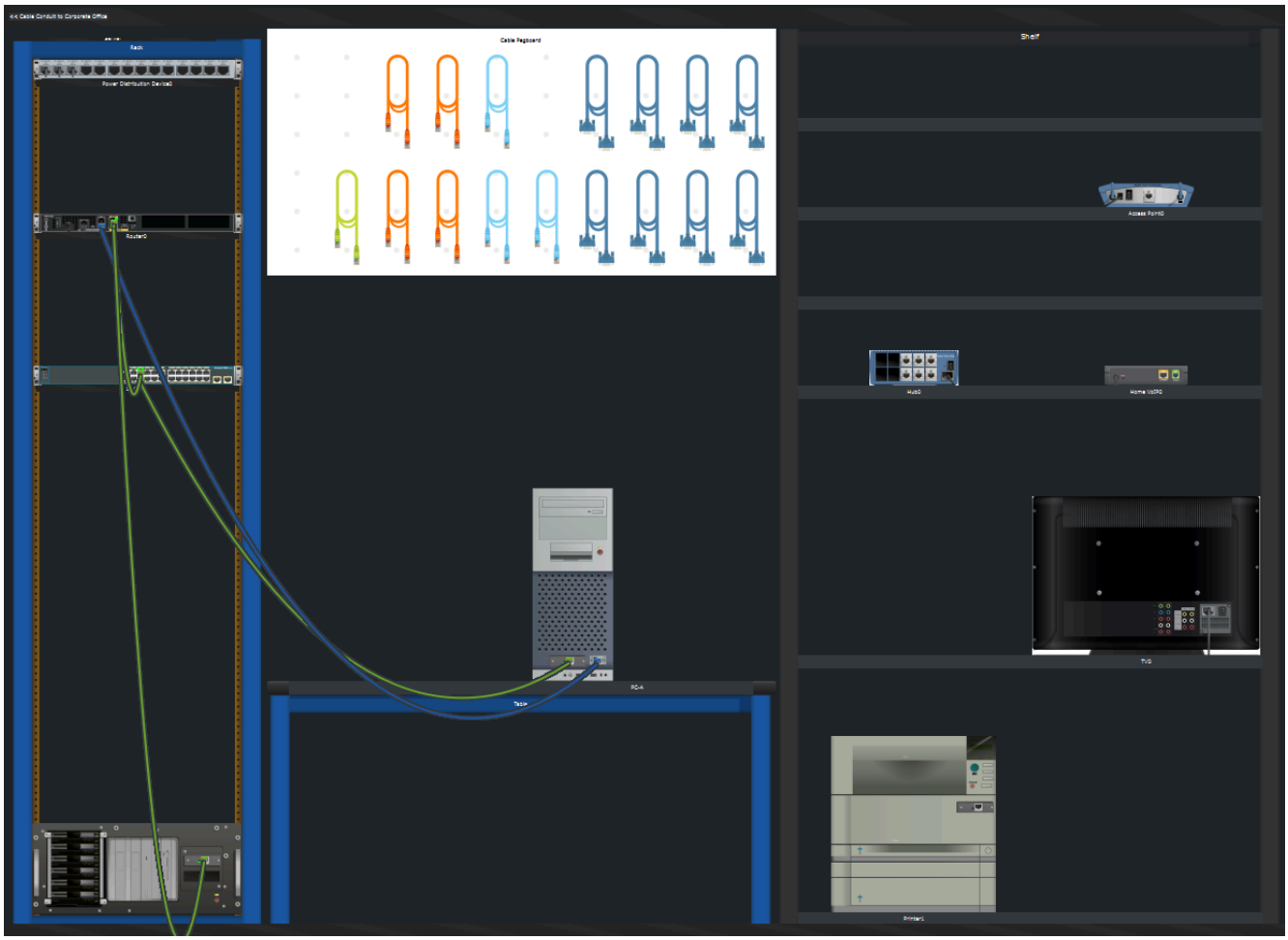
From **PC-B**, issue a ping to the **R1** default gateway link local address. Was it successful?

From **Server**, issue a ping to the **R1** IPv6 unicast address 2001:db8:acad::1. Was it successful?

Reflection Questions

1. In researching a network connectivity issue, a technician suspects that an interface was not enabled. What **show** command could the technician use to troubleshoot this issue?
2. In researching a network connectivity issue, a technician suspects that an interface was assigned an incorrect subnet mask. What **show** command could the technician use to troubleshoot this issue?





```
C:\>ping 192.168.0.10

Pinging 192.168.0.10 with 32 bytes of data:

Request timed out.
Reply from 192.168.0.10: bytes=32 time<1ms TTL=127
Reply from 192.168.0.10: bytes=32 time<1ms TTL=127
Reply from 192.168.0.10: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.0.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 2001:db8:acad::10

Pinging 2001:db8:acad::10 with 32 bytes of data:

Reply from 2001:DB8:ACAD::10: bytes=32 time=1ms TTL=127
Reply from 2001:DB8:ACAD::10: bytes=32 time<1ms TTL=127
Reply from 2001:DB8:ACAD::10: bytes=32 time<1ms TTL=127
Reply from 2001:DB8:ACAD::10: bytes=32 time=2ms TTL=127

Ping statistics for 2001:DB8:ACAD::10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 2ms, Average = 0ms
```

Reflection Questions

1. In researching a network connectivity issue, a technician suspects that an interface was not enabled. What show command could the technician use to troubleshoot this issue?

2. In researching a network connectivity issue, a technician suspects that an interface was assigned an incorrect subnet mask. What show command could the technician use to troubleshoot this issue?

ตอบ

1. คำถาม: เทคนิคเชียนสงสัยว่าอินเทอร์เฟซอาจจะยังไม่ได้เปิดใช้งาน (not enabled) ควรใช้คำสั่ง `show` ใด?

คำตอบ: > **`show ip interface brief`** (หรือ **`show ipv6 interface brief`** สำหรับ IPv6)

- **เหตุผล:** คำสั่งนี้จะแสดงสถานะโดยย่อของทุกอินเทอร์เฟซในหน้าเดียว หากอินเทอร์เฟซนั้นยังไม่ได้เปิดใช้งาน (ไม่ได้พิมพ์คำสั่ง `no shutdown`) ในคอลัมน์ **Status** จะแสดงสถานะเป็น **`administratively down`** ทำให้ตรวจสอบและตรวจพบได้อย่างรวดเร็วครับ

2. คำถาม: เทคนิคเชียนสงสัยว่าอินเทอร์เฟซอาจจะถูกตั้งค่า Subnet Mask ผิดพลาด ควรใช้คำสั่ง `show` ใด?

คำตอบ: สามารถใช้ได้ 2 คำสั่งหลักๆ ตามความเหมาะสมครับ:

- **`show run`** (หรือ **`show running-config`**):
  - **เหตุผล:** เป็นวิธีที่ง่ายและนิยมที่สุด เพราะจะแสดงโครงสร้างการตั้งค่าปัจจุบันทั้งหมด คุณสามารถเลื่อนลงไปดูที่ตัวอินเทอร์เฟซนั้นๆ เพื่อเช็คดูว่า IP Address และ Subnet Mask ที่พิมพ์ลงไปตรงกับที่วางแผนไว้หรือไม่ เช่น `ip address 192.168.1.1 255.255.255.0`
- **`show ip interface <`**ชื่ออินเทอร์เฟซ**`>`** (เช่น `show ip interface gigabitEthernet 0/0/0`):
  - **เหตุผล:** หากต้องการเจาะจงดูเฉพาะพอร์ตนั้น คำสั่งนี้จะแสดงรายละเอียดเชิงลึกของเลเยอร์ 3 รวมถึงแสดง IP Address และ Subnet Mask (ในรูปแบบตัวเลขฐานสิบหรือ Prefix) ของอินเทอร์เฟซนั้นอย่างชัดเจนครับ